2)

$$X(s) = e^{-2s}\,\frac{s+2}{s-2}$$

$$x(t) = 0 \quad \forall \; t > 2$$

$$y(t) = -\frac{2}{3}e^{2t}u(-t) + \frac{1}{3}e^{-t}u(t)$$

a) $$Y(s) = \frac{2/3}{s-2} + \frac{1/3}{s+1} = \frac{\frac{2}{3}(s+1)+\frac{1}{3}(s-2)}{(s-2)(s+1)}$$

$$H(s) = \frac{Y(s)}{X(s)} \quad \text{al ser LTI}$$

$$H(s) = \frac{\frac{2}{3}(s+1)+\frac{1}{3}(s-2)}{(s-2)(s+1)} \Big/ \frac{e^{-2s}(s+2)}{s-2} = \frac{\frac{2}{3}(s+1)(s-2)+\frac{1}{3}(s-2)^2}{e^{-2s}(s-2)(s+1)(s+2)} =$$

$$(s-2)(s+2) = s^2 + 2s - 2s - 4 = s^2 - 4$$

$$H(s) = \frac{2/3 + \frac{1/3(s-2)}{s+1}}{e^{-2s}(s+2)} = \frac{2e^{2s}}{3(s+2)} + \frac{(s-2)e^{2s}}{3(s+1)(s+2)}$$

Fracciones simples en el segundo término:

$$\frac{A}{3(s+1)} + \frac{B}{3(s+2)} = \frac{(s-2)e^{2s}}{3(s+1)(s+2)} \Rightarrow (s+2)A + 3(s+1)B = (s-2)e^{2s}$$

$$H(s) = \frac{2e^{2s}}{3(s+2)} + \frac{4e^{2s}}{3(s+2)} - \frac{e^{2s}}{(s+1)}$$

RoC de $X(s) = Re(s) < 2$

RoC de $Y(s) = Re(s) < 2 \wedge Re(s) > -1$

ROC de $H(s)$ debe contener $ROC(X(s)) \cap ROC(Y(s))$

Entonces $ROC(H(s)) = Re(s) > -1 \wedge Re(s) < 2$

Polos de $H(s) \Rightarrow p_{1,2}$ en $s=-2$ y p_3 en $s=-1$

Causal → Roc = la derecha del polo más derecho. ⇐ Es causal.

Estable → Roc incluye $Re(s)=0$ ⇐ Es estable.

Ec. en diferencias: $\frac{Y(s)}{X(s)} = H(s) \Rightarrow e^{-2s}(s+2)Y(s) = \frac{2}{3}X(s) + \frac{X(s)(s-2)}{3(s+1)}$

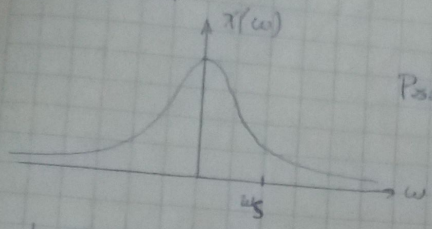
$$se^{-2s}Y(s) + 2e^{-2s}Y(s) = \frac{2}{3}X(s) + \frac{sX(s)}{3(s+1)} - \frac{2X(s)}{3(s+1)}$$

$y'(t-2)$

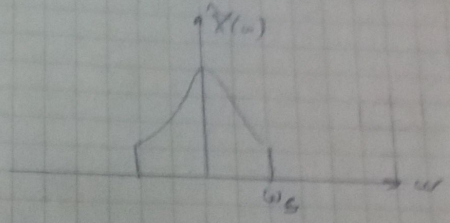
No se puede por el e^{2s} (no es racional).

26-12-13

b) Con H_1:

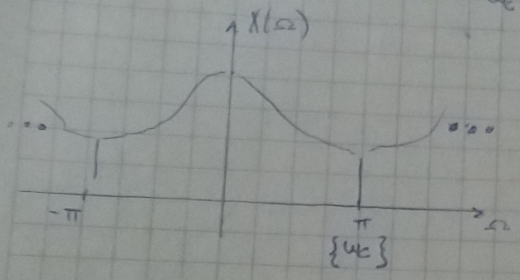


Paso por H_1 ⟹

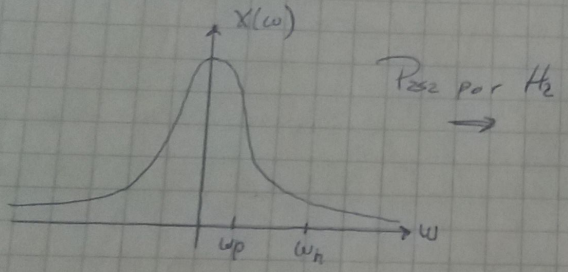


Luego de muestrear:

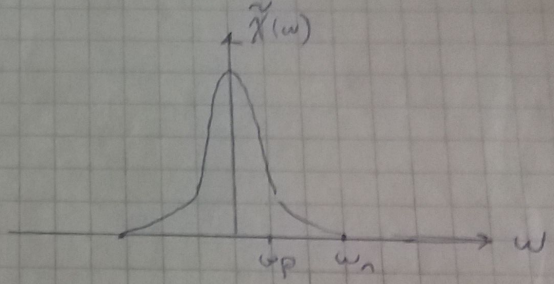
$T = \frac{\pi}{\omega_c} \Rightarrow 2\pi = \omega_s T \Rightarrow \omega_s = \frac{2\pi}{T} \Rightarrow \omega_s = 2\omega_c$



Con H_2:

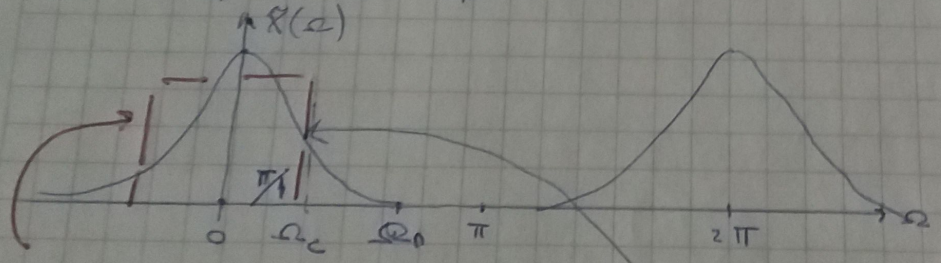


Paso por H_2 ⟹



Ahora muestreo a mayor frecuencia:

$2\pi = \omega_s \frac{T}{4} \Rightarrow \omega_s = \frac{8\pi}{T} = 8\omega_c$



$\frac{\pi}{4} = \omega_n \frac{\pi}{4\omega_c}$

$\omega_n = \omega_s$?

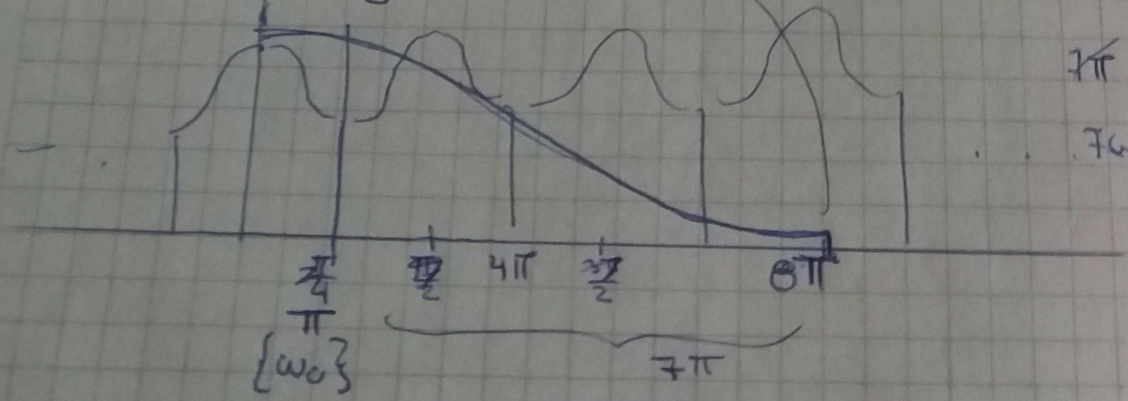
$\pi = \omega_n \frac{\pi}{4\omega_c}$

$3\omega_c = \omega_n$

$7\pi = \omega_n \frac{\pi}{\omega_c}$

$7\omega_c = \omega_n$

Filtro H_d ⟶ Luego promediar ↓4



$\omega_p = \omega_c$ ← Min ω_p

$\omega_n = \omega_n$ ← Max ω_n